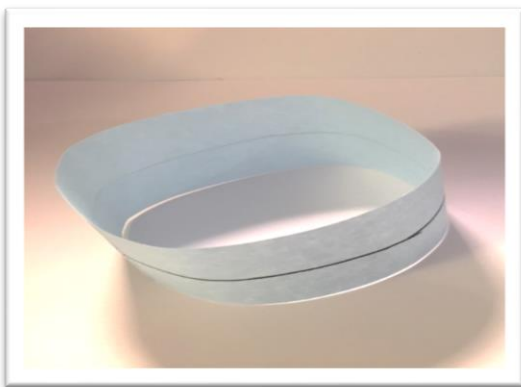


The Magical Möbius Loop!

Let's explore some interesting surfaces, which can be made from a sheet of paper.



Take a strip of paper and stick the two ends together.

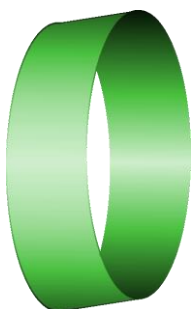
You have just created a surface called a

Cylinder

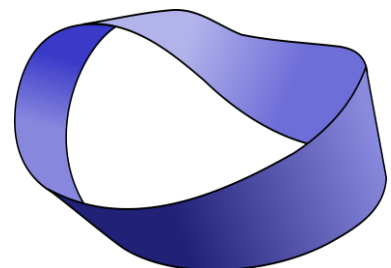
Take another strip of paper, put a twist into the strip by turning one end through half a turn, and then join the two ends together.



This surface is called a
Möbius Loop



What differences are there between these two surfaces?



Can you **predict** what happens next in the following experiments?



1. Draw a line along the centre of both the cylinder & the Möbius loop.

What do you discover?

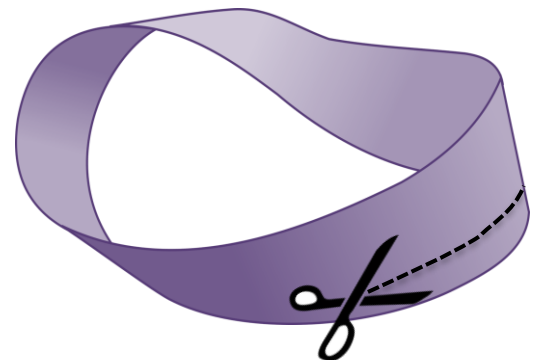


2. Now cut along this line - cut both the **cylinder** and the **Möbius loop** in half.

Try to guess what will happen first!

3. Make another **Möbius loop**.

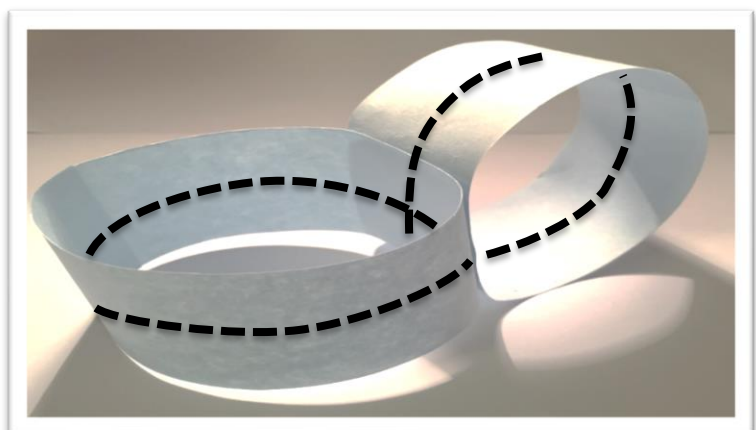
Last time, you cut it in the middle. This time, cut it **closer to one of the edges**, about $\frac{1}{3}$ of the way from the edge of the loop.



Is the result the same as before?

4. Make **two cylinders** and stick them together at right angles – it should look like a twisted figure 8. What do you think you will get if you cut both strips down the middle, along their length?

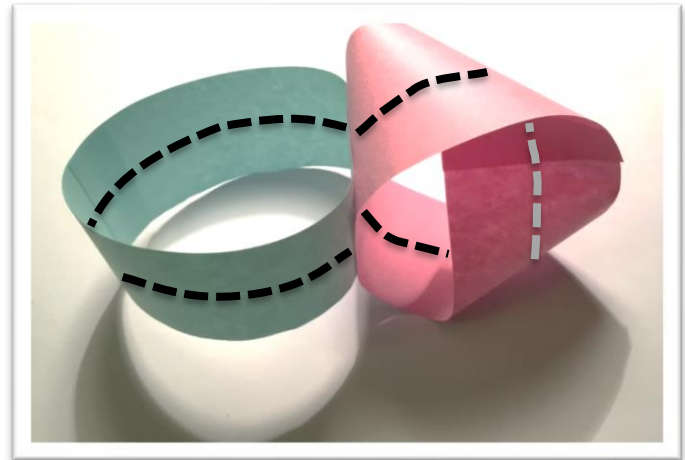
Let's try it! Cut down the middle of each loop - you will need to make two cuts to complete this step.



5. Make **one Möbius loop** and **one cylinder** and stick them together right angles.

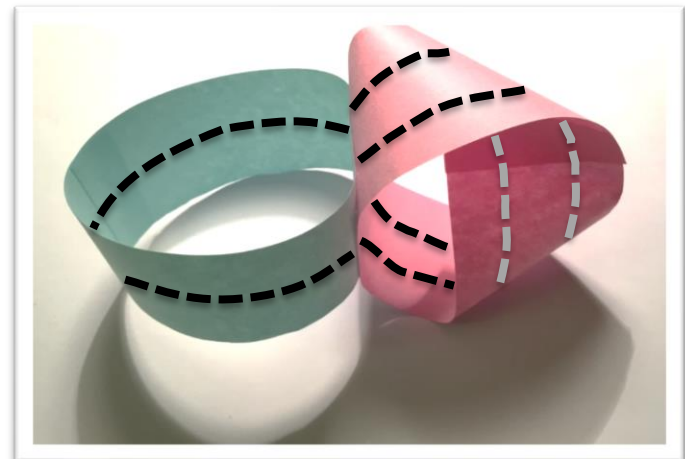
Now cut them both down the middle - you will need to make two cuts to complete this step.

Before you start cutting, don't forget to guess what shape you think you will get!

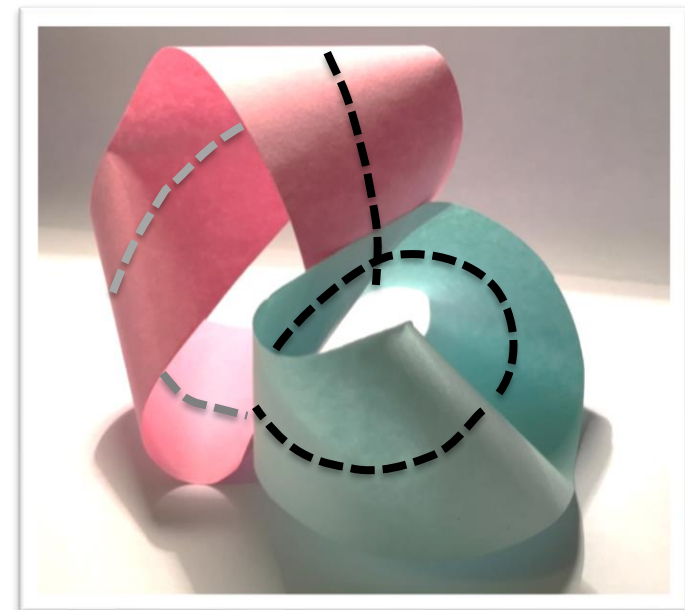


6. Follow the instructions for part 5 as above, but this time cut the **Möbius loop** near to the edge (as you did in step 3), a third of the way along the width of the strip.

Then cut the cylinder along the middle as before.



7. Make **two Möbius loops**, but twist the two loops in opposite directions - one loop should spiral one way, and the other loop should spiral the opposite way. Now stick the loops together at right angles, and cut along the middle of the strips, cutting through where the two strips are stuck together.



In these exercises, you have been exploring the **mathematics of shapes**, which is called **topology**!